



AI-Driven Enterprise Knowledge Digital Twin using Agentic AI and Hybrid RAG

Major Project Review - 1

Batch ID: AIML-B02

Project Guide

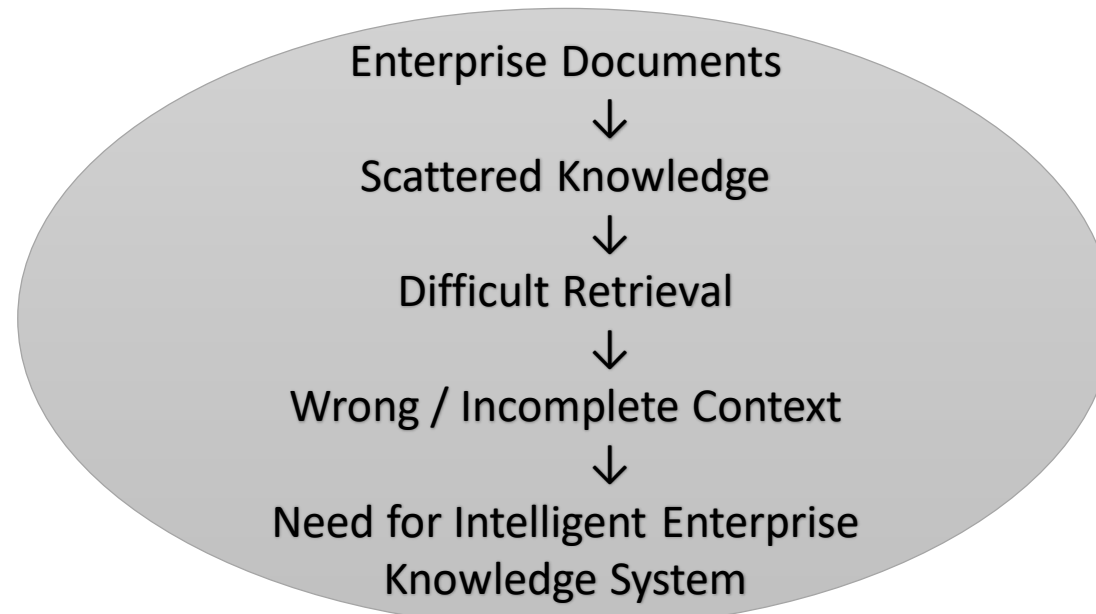
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Problem statement

- Enterprise knowledge is distributed across **PDFs, Word files, manuals, reports and internal documents**.
- Conventional keyword search may fail to understand the **semantic meaning** of user queries.
- Standard RAG can retrieve a limited set of chunks and may miss relevant information.
- Employees may have different **roles and document-access permissions**.
- Complex questions may require information from **multiple documents**.
- Therefore, an intelligent system is required to **retrieve, reason, validate and present enterprise knowledge securely**.



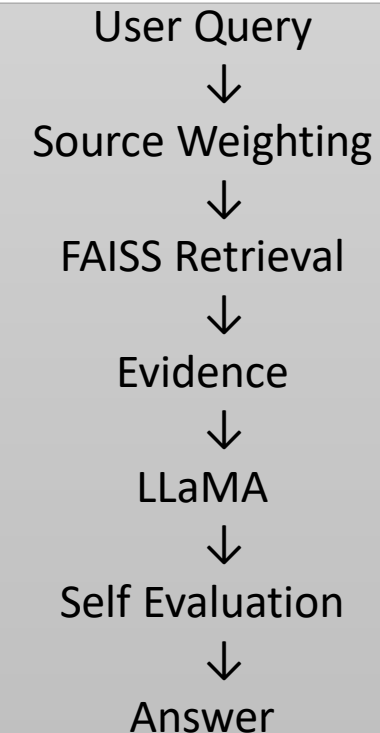
Literature Review

Ref 01 – Agentic AI-Driven Technical Troubleshooting for Enterprise Systems: A Novel Weighted Retrieval-Augmented Generation Paradigm

Rajat Khand , 2024 | arXiv

Proposed Method | Data & Setup | key Results | Strengths & Limitations | Relevance to my research |

- Collects knowledge from **multiple enterprise sources**.
- Documents are converted into **chunks and embeddings**.
- Uses **FAISS** for dense retrieval.
- Assigns **dynamic weights** based on query context.
- Retrieved evidence is passed to **LLaMA for answer generation and self-evaluation**.



CORE IDEA:

Query-dependent weighted retrieval + self-evaluation

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Implementation Tech Stack

Component	Technology
Language	Python
Retrieval	FAISS
Embeddings	Dense vector embeddings
Generation	LLaMA
Architecture	Weighted RAG
Evaluation	LLaMA-based self-evaluator

Official GitHub: Not publicly verified

Live Demo: Not publicly verified

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Enterprise Knowledge Sources

- 1,200 Product Manuals
- 40,000 FAQs
- Troubleshooting Guides
- Internal Knowledge Bases

Processing :

Chunking → MiniLM Embeddings → Separate FAISS Indexes

Software :

Python 3.10 | PyTorch 2.0 | FAISS 1.7.3 | Transformers

Models :

all-MiniLM-L6-v2 → Embedding

LLaMA 3.1 70B → Generation + Self-Evaluation

Hardware :

4 × NVIDIA A100 (80 GB) | 2 × 40-core Intel Xeon

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- Preliminary evaluations show improved **troubleshooting accuracy**.
- The approach aims to reduce **resolution time**.
- Dynamic weighting helps adapt retrieval to different technical queries.
- Self-evaluation improves contextual reliability.
- The framework is designed for heterogeneous enterprise knowledge sources.

Better source selection → better evidence → more reliable enterprise answers

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Strengths

- Handles **multiple enterprise knowledge sources**
- Query-dependent source weighting
- Dense vector retrieval using FAISS
- Self-evaluation of generated answers
- Designed for complex technical troubleshooting

Limitations

- Primarily focuses on **technical troubleshooting**
- Depends heavily on retrieval quality
- Dynamic weighting can add system complexity
- Conversational AI and reinforcement-learning-based weighting are described as future directions.

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From this paper → Our project

Paper provides	Our implementation
Weighted Retrieval	Hybrid Retrieval
Multiple sources	Enterprise Documents
FAISS retrieval	Vector Retrieval
Self-evaluation	Answer validation
Agentic direction	Agentic AI

Our additions
RBAC + Organization/Department access + Citations + Digital Twin

Literature Review **Ref 02** – Benchmarking Deep Search over Heterogeneous Enterprise Data Prafulla Kumar Choubey et al. | EMNLP 2025

| Proposed Method | Data & Setup | key Results | Strengths & Limitations | Relevance to my research |

- Introduces **HERB — Heterogeneous Enterprise RAG Benchmark**.
- Evaluates Deep Search over diverse enterprise sources.
- Sources include **documents, meeting transcripts, Slack, GitHub and URLs**.
- Tests **source-aware, multi-hop retrieval**.
- Includes both answerable and unanswerable queries.
- Focuses on whether RAG systems can retrieve all required evidence.

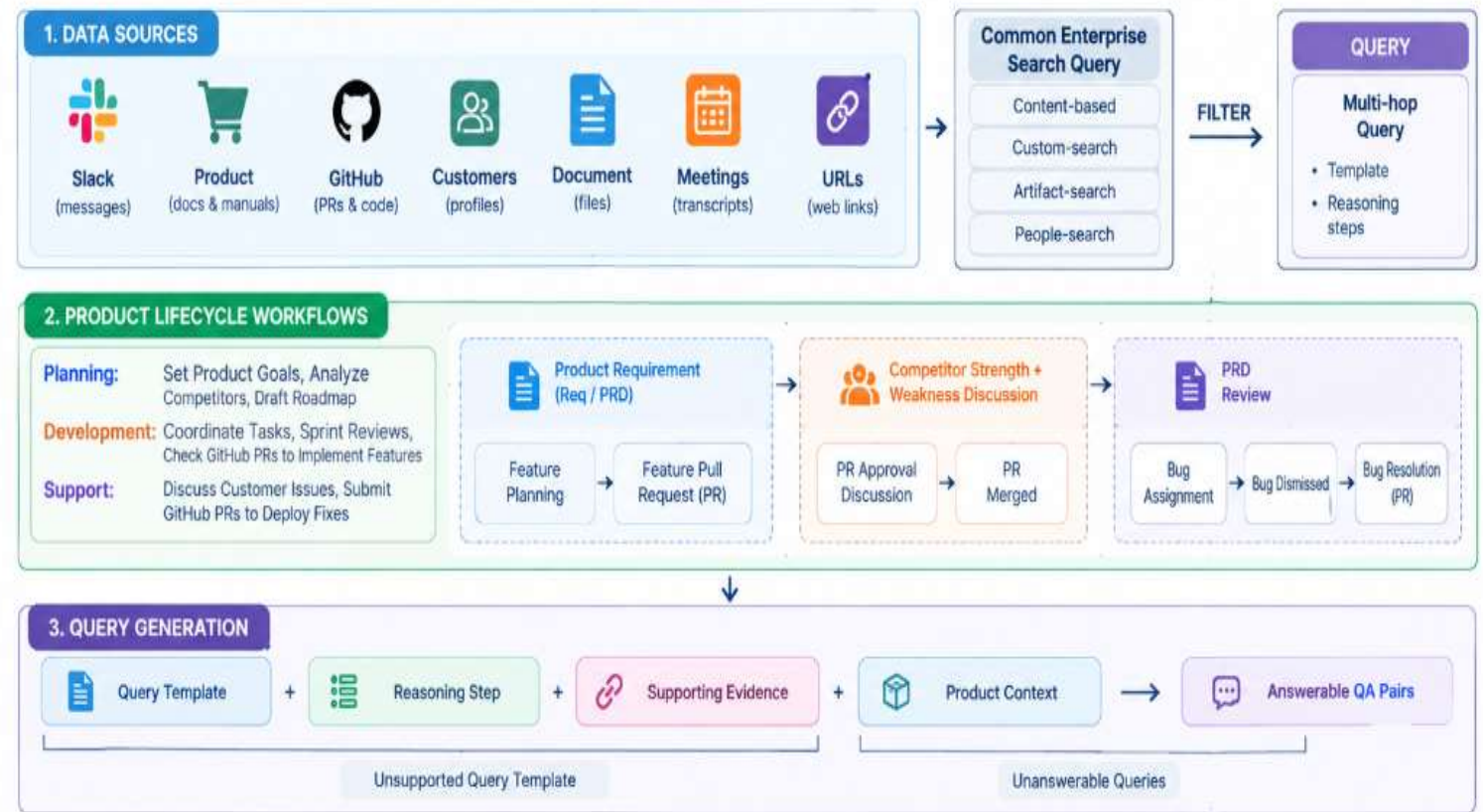


Figure 1: HERB Workflow-Guided Data Synthesis Framework

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Implementation TechStack

Component	Technology used
Programming	Python
Embedding Model	OpenAI text-embedding-3-large
LLM for generation	GPT-4o
Agent Framework	ReAct
Retrieval	Dense + Sparse Hybrid Retrieval
Graph-based RAG	GraphRAG, HippoRAG 2, Proposition-Graph RAG
Structured Search	8 search tools
Evaluation	RAG performance + long-context LLM evaluation

GitHub: <https://github.com/SalesforceAIResearch/HERB>

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Dataset Size

- 39,190 enterprise artifacts

Sources

- Documents
- Meeting transcripts
- Slack messages
- GitHub
- URLs

Queries

- Answerable + Unanswerable

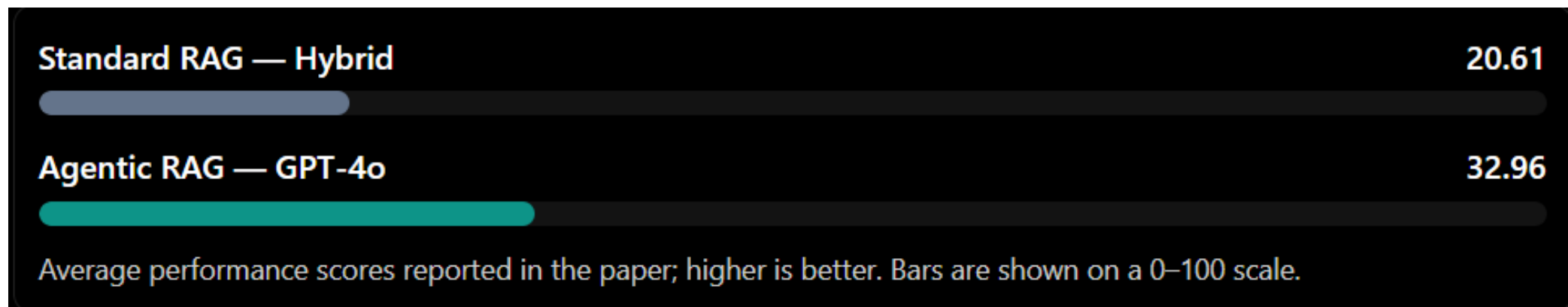
Data generation

- The benchmark uses a **synthetic enterprise environment** simulating business workflows across product planning, development and support.

Literature Review

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Main Findings :

- Agentic RAG outperformed standard RAG in the benchmark.
- The best-performing agentic configuration achieved an average score of 32.96.
- The strongest standard RAG baseline, Hybrid, achieved 20.61.
- Even agentic systems struggled to retrieve all required evidence.
- The authors identify retrieval as a major bottleneck.

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Strengths

- Multiple enterprise data sources
- Complex multi-hop queries
- Answerable and unanswerable questions
- Public benchmark and code
- Supports detailed RAG evaluation

Limitations

- Uses a synthetic enterprise environment
- Retrieval remains challenging
- Complex queries need evidence from multiple sources
- Benchmark itself is not a complete enterprise application

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What We Take from HERB

Multi-source knowledge retrieval

Support for retrieving information from multiple enterprise document types.

Complex query handling

Motivates retrieving and connecting evidence from multiple documents.

Retrieval evaluation

Motivates testing retrieval quality instead of checking only generated answers.

Our Extension:

HERB's multi-source retrieval problem → Our Hybrid RAG + Agentic AI + RBAC + Citations

Literature Review **Ref 03** – AgenticRAG: Agentic Retrieval for Enterprise Knowledge Bases Susheel Suresh et al. | 2026 | arXiv

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Fig 1. Agentic Loop

Instead of:

Query → Retrieve Once → Answer

the system performs:

Query → Search → Find → Open → Summarize → Answer

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Implementation Tech Stack

Layer	Technology / Approach
LLM	Large Language Model
Retrieval	Agentic / Iterative Retrieval
Agent Tools	Search, Find, Open
Knowledge Base	Enterprise Knowledge Bases
Generation	Retrieval-Augmented Generation
Evaluation	BRIGHT, WixQA, FinanceBench

Literature Review

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This paper doesn't use one normal dataset with columns like:

id | name | department | salary

Instead, it evaluates the retrieval system on **three benchmarks**:

1. BRIGHT

Long-context retrieval

Metric:

Recall@1

2. WixQA

Enterprise question answering

Metric:

Factuality

3. FinanceBench

Financial document question answering

Metric:

Answer Correctness

The paper reports results across all three benchmarks.

Literature Review

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BRIGHT

49.6% Recall@1

+21.8 percentage points over the best embedding baseline.

WixQA

0.96 Factuality

+13% relative improvement

FinanceBench

92% Answer Correctness

Within 2 percentage points of oracle access to the true evidence.

Ablation finding

5.9× improvement

Literature Review

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Strengths

- Iterative retrieval
- LLM-driven tool selection
- Search + in-document navigation
- Works with existing enterprise search infrastructure
- Produces cited answers
- Evaluated on multiple enterprise-oriented benchmarks

Limitations

- More complex than single-shot RAG
- Multiple tool calls can increase latency and token usage
- Depends on the underlying search infrastructure
- Does not provide our project's **RBAC + organization/departement security model**
- Does not implement our complete **Enterprise Knowledge Digital Twin**

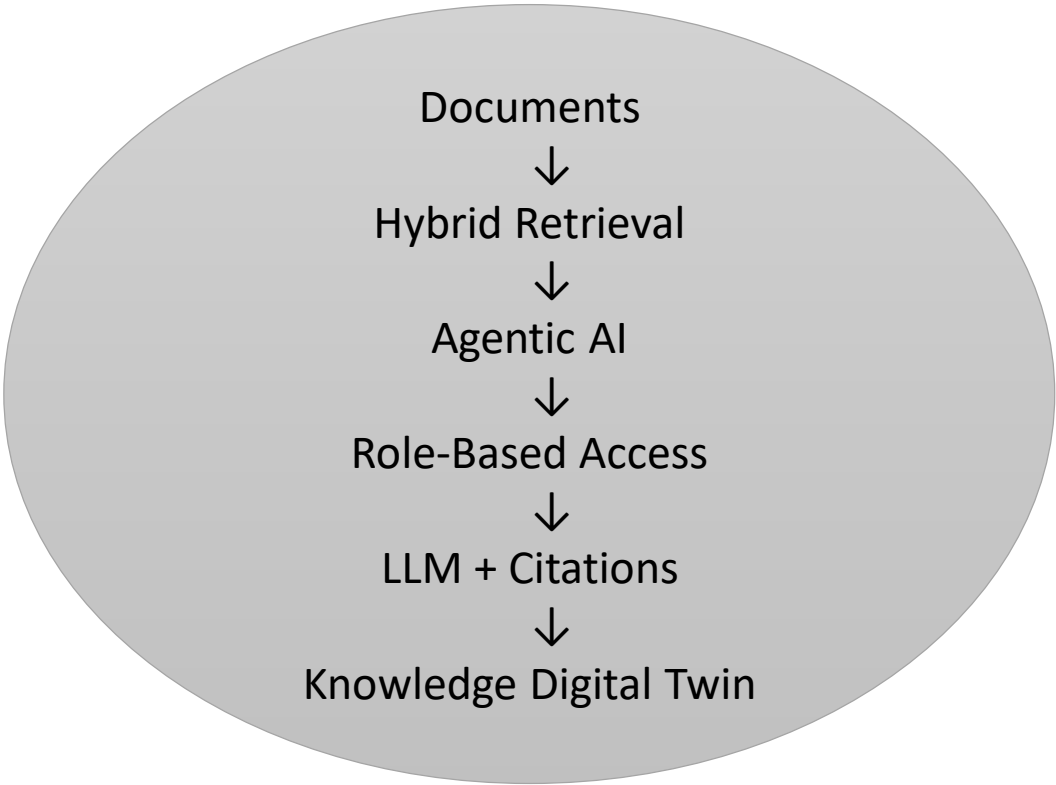
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From AgenticRAG



We implement



Literature Review - Summary

Reference	Year	Main Idea	Strengths	Limitations
Khanda – Agentic AI-Driven Technical Troubleshooting for Enterprise Systems	2024	Agentic weighted RAG for enterprise technical troubleshooting using multiple knowledge sources	Query-dependent retrieval, enterprise-focused, agentic reasoning	Preliminary evaluation; mainly troubleshooting-oriented
Raina & Gales – Question-Based Retrieval using Atomic Units for Enterprise RAG	2024	Improves RAG retrieval by decomposing chunks into atomic statements and generating synthetic questions	Higher retrieval recall, better question–document matching	Focuses mainly on retrieval; no agentic workflow or access control
Choubey et al. – Benchmarking Deep Search over Heterogeneous Enterprise Data (HERB)	2025	Benchmarks deep search over heterogeneous enterprise data such as documents, Slack, GitHub and URLs	Realistic enterprise setting, answerable & unanswerable queries, agentic evaluation	Mainly a benchmark; retrieval remains a major bottleneck
Suresh et al. – AgenticRAG: Agentic Retrieval for Enterprise Knowledge Bases	2026	Uses a reasoning LLM with Search, Find, Open and Summarize tools for iterative enterprise retrieval	Iterative retrieval, multi-step reasoning, cited answers, strong benchmark results	Higher complexity/latency; depends on existing search infrastructure

Review – 1

Questions?